SELF TEST

1. 50 grams of ice is mixed with 40 gm of water at 60°C. Then the ratio of ice and water at equilibrium is

   1) 2 : 7 2) 7 : 2 3) 2 : 3 4) 2 : 5

2. A lead bullet just melts when stopped by an obstacle. Assuming that 25% of the heat is absorbed by the obstacle, find the velocity of the bullet if its initial temperature is 27°C. Given melting point of lead = 327°C, specific heat of lead = 0.03 cal/g/°C. Latent heat of fusion of lead = 6 cal/g, J = 4.2 J/cal.

   1) 23 m/s 2) 25 m/s 3) 26 m/s 4) 28 m/s

3. A large amount of ice at 0^0 C is added to a beaker of water at 30^0 C. Some of the ice melts and the rest floats on water in the beaker. The temperature of the water in the beaker is

   1) 0^0C 2) 20^0C 3) 30^0C 4) 40^0C

4. A calorimeter of water equivalent 100 grams contains 200 grams of water at 10°C. A solid of mass 500 grams at 45°C is added to the Calorimeter. If equilibrium temperature is 25°C then, the specific heat of the solid is

   1) 0.45 cal/g/°C 2) 0.1 cal/g/°C 3) 4.5 cal/g/°C 4) 0.01 cal/g/°C

5. A calorimeter of water equivalent 50 grams contains 350 g of water at 40°C. Then 100 grams of ice is added to it then resultant temperature is

   1) 0°C 2) 25°C 3) 16°C 4) 10°C

6. Boiling water is changing into steam. Under this condition, the specific heat of water is -

   1) one 2) zero 3) ∞ 4) < 1

7. The temperature of ice is –10°C [specific heat = 0.5 kcal/ (kg – °C)] and that of water 60°C. They are mixed in equal amounts. What part of the ice will be melted -

   1)(5/6) th 2)(11/16) th
   3)whole of it will be melted 4)(5/11) th

8. Melting point of ice

   1)increases with increasing pressure
   2)decreases with increasing pres sure
   3)in independent of pressure 4)is proportional to pressure

9. 5 g of steam at 100°C is passed into 6 gm of ice at 0°C. If the latent heats of steam and ice are 540 cal/gm and 80 cal/gm, then the final temperature is -

   1)0°C 2) 50°C 3) 30°C 4) 100°C

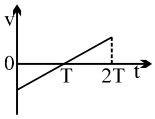
10. In above question the mixture contains

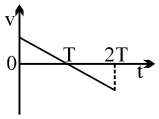
    1) 11 gm of water 2) 8 gm of water
    3)6 gm of water 4) 5 gm of water

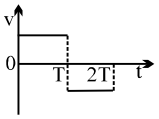
Q: 11-14

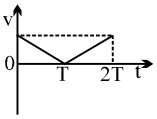
The x-t graph of a particle moving along a straight line is shown in figure

11 . The v-t graph of the particle is correctly shown by

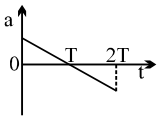
1) 

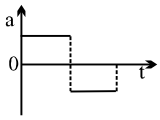
2) 

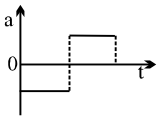
3) 

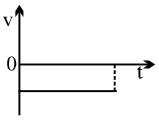
4) 

12. The a-t graph of the particle is correctly shown by

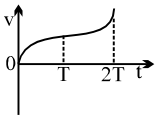
1) 

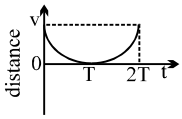
2) 

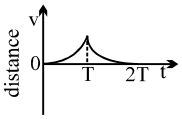
3) 

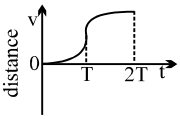
4) 

13. The distance-time graph of the particle is correctly shown by

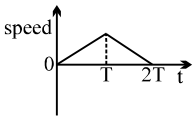
1) 

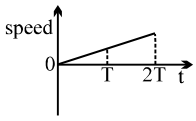
2) 

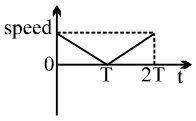
3) 

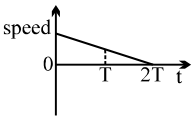
4) 

14. The speed-time graph of the particle is correctly shown by

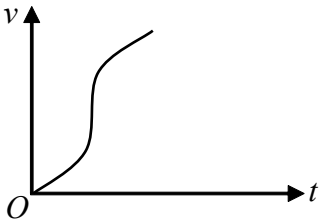
1) 

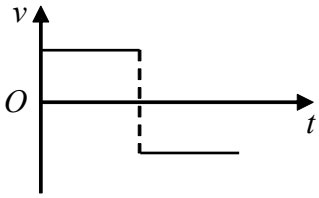
2) 

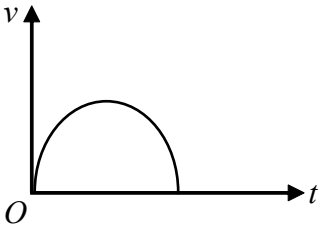
3) 

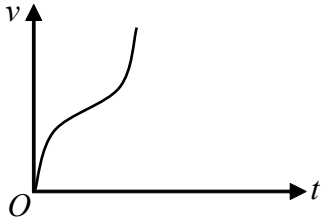
4) 

15. The acceleration versus time graph of a particle is as shown in figure. The respective v – t graph of the particle is

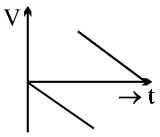
1) 

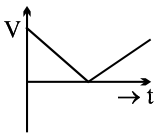
2) 

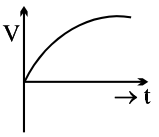
3) 

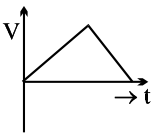
4) 

16. The velocity- time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?

1) 

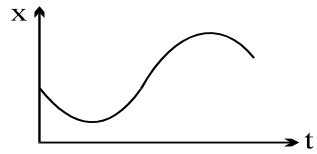
2) 

3) 

4) 

17. The velocity- time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?

13. The graph of position x versus time t represents the motion of a particle. If b and c are both positive

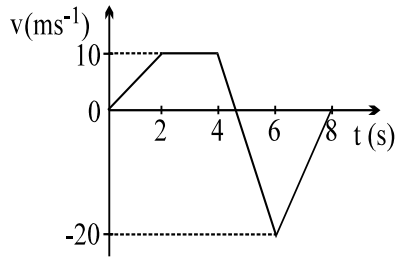


constants, which of the following expressions best describes the acceleration ' a ' of the particle?

1) a = b – ct
2) a = + b
3) a = – c
4) a = b + ct

Q :14 -19

The figure shows a velocity-time graph of a particle moving along a straight line



14. Choose the incorrect statement. The particle comes to rest at

1) t = 0 s
2) t = 5 s
3) t = 8 s
4) none of these

15. Identify the region in which the rate of change of velocity $\left|\frac{\Delta\vec{v}}{\Delta t}\right|$ of the particle is maximum

1) 0 to 2s
2) 2 to 4s
3) 4 to 6 s
(4) 6 to 8 s

16. If the particle starts from the position x_0 = –15 m, then its position at t = 2s will be

1) –5 m
2) 5 m
3) 10 m
4) 15 m

17. The maximum displacement of the particle is

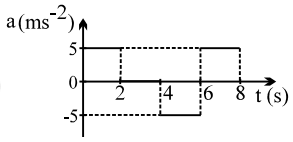
1) 35 m
2) 23.3 m
3) 18.3 m
4) zero

18. The total distance travelled by the particle is

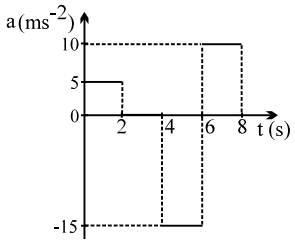
1) 66.6 m
2) 51.6 m
3) zero
4) 36.6 m

19. The correct acceleration-time graph of the particle is shown as

1)



2)



SELF TEST

1. A vector having unit magnitude is called
   1) zero 2) proper 3) co-planar 4) unit vector
2. Distance is a
   1) Scalar quantity 2) vector 3) both (1) and (2) 4) None
3. A vector of magnitude 6 units along negative y axis is
   1) $6\hat{j}$ 2) $-6\hat{i}$ 3) $-6\hat{j}$ 4) $6\hat{k}$
4. A vector of magnitude5 units in **xz** plane is
   1) $3\hat{i} + 4\hat{k}$ 2) $4\hat{j} + 3\hat{k}$ 3) $5\hat{i}$ 4) $4\hat{j} + 3\hat{i}$
5. What is the condition for the vectors 2i+3j – 4k and 3i – aj + bk to be parallel ?
   1) a = –9/2, b = – 62) a = –6, b = –9/2 3) a = 4, b = 5 4) a = 8, b = 2
6. Two vectors are given by $\bar{a}$ = -2$\hat{i}$ + j-3k and $\bar{b}$ = 5$\hat{i}$ +3j-2k. Third vector $\bar{c}$ is if 3a + 2b – c = 0
   1) 4i + 9j – 13 k 2) –4i – 9j + 13 k 3) 4i–9j–13k 4) none
7. The minimum number of unequal forces in a plane that can keep a particle in equilibrium is
   1) 4 2) 2 3) 3 4) 6
8. Which of the following sets of forces acting simultaneously on a particle keep it in equilibrium?
   1) 3N, 5N, 10N 2) 4N, 7N, 12N 3) 2N, 6N, 5N 4) 5N, 8N, 1N
9. The maximum and minimum resultants of two forces are in the ratio 4:3. The forces are in the ratio
   1) 7:1 2) 1:5 3) 4:7 4) 3:7
10. A is a vector which when added to the resultant of vectors $\left(2\hat{i}-3\hat{j}+4\hat{k}\right)$ and $\left(\hat{i}+5\hat{j}+2\hat{k}\right)$ yields a unit vector along the y-axis. Then vector A is
    1) $-3\hat{i}-\hat{j}-6\hat{k}$ 2) $3\hat{i}+\hat{j}-6\hat{k}$ 3) $3\hat{i}-\hat{j}+6\hat{k}$ 4) $3\hat{i}+\hat{j}+6\hat{k}$
11. One of the rectangular components of a velocity of 20 ms^{-1} is 10 ms^{-1} . Find the other component.
    1) 10 $\sqrt{3}\,ms^{-1}$ 2) 20 $\sqrt{3}\,ms^{-1}$ 3) 5 $\sqrt{3}\,ms^{-1}$ 4) 35 $\sqrt{3}\,ms^{-1}$
12. Angle (in rad) made by the vector $\sqrt{3}\hat{i}+\hat{j}$ with the x-axis is
    (1) $\pi/6$ (2) $\pi/4$ (3) $\pi/3$ (4) $\pi/4$
13. If $\vec{A}$ =2i-3j+4k, its components in yz plane and zx plane are respectively
    1) $\sqrt{13}$ and 5 2) 5 and $2\sqrt{5}$ 3) $2\sqrt{5}$ and $\sqrt{13}$ 4) $\sqrt{13}$ and $\sqrt{29}$
14. Two vectors a and b have equal magnitudes of 12 units. These vectors are making angles 30^0 and 120^0 with the x axis respectively. Their sum is $\vec{r}$. Find the x and y components of $\vec{r}$.

11. $\int x + x^2\, dx =$

1) $x^2 + \frac{x^3}{3} + c$ 2) $\frac{x^2}{2} + \frac{x^3}{3} + c$ 3) $2x^2 + \frac{x^3}{3} + c$ 4) $\frac{x^2}{2} + x^3 + c$

12. $\int (x + Sinx)\, dx =$

1) $\frac{x^2}{2} - Cosx + c$ 2) $\frac{x^2}{2} - 2Cosx + c$ 3) $\frac{x^2}{2} - Sinx + c$ 4) $\frac{x^2}{2} + Cosx + c$

13. $\int (x + \frac{1}{x})\, dx =$

1) x + logx +c 2) 2x + logx + c 3) $\frac{x^2}{2} + \log x + c$ 4) $\frac{x^2}{2} - logx + c$

14. $\int (x + 2x^3)\, dx =$

1) $x + \frac{2x^4}{4} + c$ 2) $x + \frac{x^4}{4} + c$ 3) $\frac{x^2}{2} + \frac{2x^4}{4} + c$ 4) $\frac{x^2}{2} + \frac{x^4}{4} + c$

15. $\int (3 + 2x)\, dx =$

1) $x + \frac{x^2}{2} + c$ 2) $3x + \frac{x^2}{2} + c$ 3) $3x + x^2 + c$ 4) $x + x^2 + c$

16. $\int (1 - \sec^2 x)\, dx =$

1) $x - \tan x + c$ 2) $x + \tan x + c$ 3) $2x + \tan x + c$ 4) $3x + \tan x + c$

17. $\int \left(a - a^{nx}\right) dx =$

1) $ax - \frac{a^{nx}}{n \log a} + c$ 2) $ax + \frac{a^{nx}}{n \log a} + c$ 3) $\frac{ax - a^{nn}}{n \log a}$ +C 4) $\frac{ax + a^{nx}}{n \log a}$ +C

18. $\int Cot^2 x\, dx =$

1) -Cot *x* - *x* + c 2) Tan *x* - *x* + c 3) Cot *x* - *x* + c 4) Sin *x* - *x* + c

19. $\int \frac{ax + b}{cx + d} dx =$

1) $\frac{ax}{c} - \frac{bc - ad}{c^2} \log |cx + d| + c$ 2) $\frac{ax}{c} + \frac{bc - ad}{c^2} \log |cx + d| + c$

3) $\log \left| \frac{ax + b}{cx + d} \right| + c$ 4) $\frac{ax}{c} + \frac{b}{c} \log |cx + d| + c$

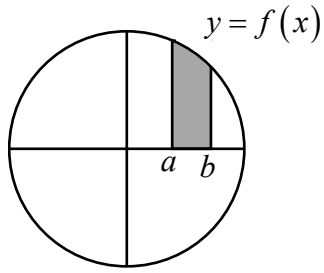
20. $\int \frac{Sin^2 x}{1 - Cosx} dx =$

1) x Sin x + c 2) x - Sin x + c 3) x + Sin x + c 4) x^2 sin x + c

KEY SHEET-1

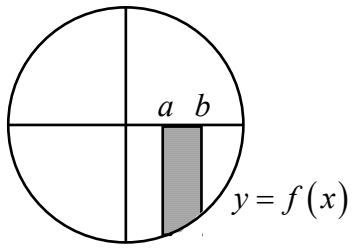
1. 4	2. 3	3. 2	4. 2	5. 2	6. 1	7. 2	8. 2	9. 1	10. 2
11. 1	12. 1	13. 3	14. 3	15. 3	16. 1	17. 1	18. 1	19. 2	20. 3

20. Shaded area represents



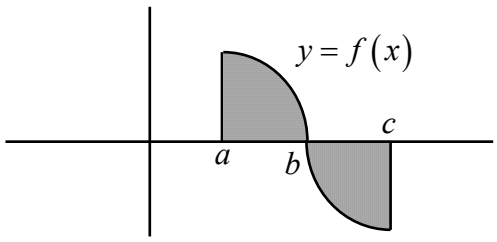
1) $\int_a^b f(x)dx$ 2) $\int_b^a f(x)dx$ 3) $2\int_a^b f(x)dx$ 4) $-2\int_a^b f(x)dx$

21. Shaded area represents



1) $\int_a^b -f(x)dx$ 2) $\int_b^a f(x)dx$ 3) $2\int_a^b f(x)dx$ 4) $2\int_b^a f(x)dx$

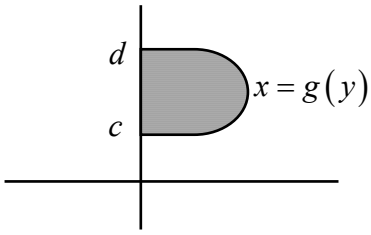
22. Shaded area represents



1) $\int_a^b f(x)dx + \int_b^c f(x)dx$ 2) $\int_a^b f(x)dx + \int_b^c -f(x)dx$

3) $\int_a^b -f(x)dx + \int_b^c f(x)dx$ 4) $\int_a^c f(x)dx$

23. Shaded area represents



1) $A = \int_c^d g(y)dy$ 2) $A = \int_c^d -g(y)dy$ 3) $A = 2\int_c^d g(y)dy$ 4) $A = 2\int_d^c g(y)dy$

11. $\frac{d}{dx}\left[\left(4x^3+1\right)\left(x^2-3\right)\right]=$

1) $20x^3-36x^2+2x$ 2) $20x^4-36x^3+2x$
3) $20x^3+36x^2+2x$ 4) $20x^4-36x^2+2x$

12. What is the derivative of $\frac{a-b\cos x}{a+b\cos x}$, where a , b are constants.

1) $\frac{2ab\cos x}{(a+b\cos x)^2}$ 2) $\frac{2ab\cos x}{(a-b\cos x)^2}$ 3) $\frac{2ab\sin x}{(a+b\cos x)^2}$ 4) $\frac{2ab}{(a+b\cos x)^2}$

13. $\frac{d}{dx}\left[\frac{3x+4}{4x+5}\right]=$

1) $\frac{-1}{(4x+5)^2}$ 2) $\frac{1}{(4x+5)^2}$ 3) $\frac{-1}{(4x-5)^2}$ 4) $\frac{1}{(4x-5)}$

14. $\frac{d}{dx}(\text{Sin}2x)=$

1)Cos2x 2)2Cos2x 3)-Sin2x 4)-2Sin2x

15. $\frac{d}{dx}(\text{Cos}(\log x))=$

1) $\frac{1}{x}$ 2) $\frac{-Sin(\log x)}{x}$ 3) $\frac{Cos(\log x)}{x}$ 4) $\frac{Sin(\log x)}{x}$

16. $\frac{d}{dx}(\text{Tan}2x)=$

1) Sec^2x 2) $\text{Sec}^2 2x$ 3)2 $\text{Sec}^2 2x$ 4)4 $\text{Sec}^2 2x$

17 $\frac{d}{dx}(\text{Sec}3x)=$

1) Sec3xTan3x 2)3 Sec3xTan3x 3)2 Sec3xTan3x 4)4 Sec3xTan3x

18. $\frac{d}{dx}(\text{Sin}^2x)=$

1)Sin2x 2)Cos2x 3)SinxCosx 4) Cos^2x

19. $\frac{d}{dx}(\log^2x)=$

1) $\frac{\log x}{x}$ 2) $\frac{2\log x}{x}$ 3) $\frac{3\log x}{x}$ 4) $\left(\frac{1}{x}\right)^2$

20. If $x = at^2$, $y = 2at$ then $\frac{dy}{dx}=$

1) 1/t 2) t 3) t^2 4) 1

21. If x = tan t, y = cos t, $0<t<\pi/2$ then $\frac{dy}{dx}=$

1) xy^2 2) $-x^2y$ 3) xy 4) $-xy^3$

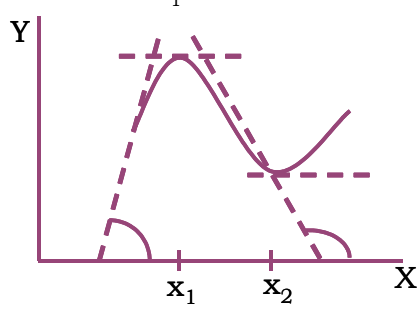
6.APPLICATION OF DERIVATIVE-I

Slope of tangent , increasing and decreasing function

1. Slope of tangent to the curve y = f(x) at the point $P(x_1, y_1)$ is $\left(\frac{dy}{dx}\right)_{(x_1, y_1)}$
2. If a line makes an angle more than right angle with the x-axis in anti clock wise direction then slope of a line is negative.
3. If a line makes an angle less than right angle with the x-axis in anti clock wise direction then slope of a line is positive.
4. If there exists a number D , such that f (c - h) > f (c) > f (c + h) for all h satisfying 0 < h < D then the function ' f ' is said to be decreasing at x = c
5. If there exists a number , such that f (c - h) < f (c) < f (c) + h) for all h satisfying 0 < h < D then the function ' f ' is said to increasing at x = c.
6. The function y = f (x) is increasing at x = c if i.e. f ' (c) > 0 and y = f (x) is decreasing at x = c if i.e. f ' (c) < 0.
7. **STATIONARY POINT:** If $f^1(a)=0$ then y = f(x) is said to be stationary at x = a. f(a) is called the stationary value of f at x = a then (a, f(a)) is called a stationary point of ' f '.

Maxima And Minima

Suppose a quantity y depends on another quantity x in a manner shown in figure. It becomes maximum at x_1 and minimum at x_2.



At these points the tangent to the curve is parallel to the x – axis and hence its slope is $\tan\theta = 0$. But the slope of the curve y(x) equals the rate of change $\frac{dy}{dx}$.Thus, at a maximum or a minimum, $\frac{dy}{dx} = 0$.

Just before the maximum the slope is positive, at the maximum it is zero and just after the maximum it is negative. Thus, $\frac{dy}{dx}$ decreases at a maximum and hence the rate of change of $\frac{dy}{dx}$ is negative at a maximum i.e.

and hence the rate of change of $\frac{dy}{dx}$ is negative at a maximum i.e.

$\frac{d}{dx}\left(\frac{dy}{dx}\right) < 0$ at a maximum. The quantity $\frac{d}{dx}\left(\frac{dy}{dx}\right)$ is the rate of change of the

ASSIGNMENT

1. A particle moves along x-axis is such a way that its coordinate (x) varies with time t, according to the expression, $x = 2 - 5t + 6t^2$. The initial velocity of the particle is
   1) –12m/s 2) 12 m/s 3) –5m/s 4) 5 m/s
2. If the displacement of a body along x-axis is given by $x = 3 + 8t + 7t^2$. Then the acceleration of the body at t = 2s is
   1) $22m/s^2$ 2) $14m/s^2$ 3) $8m/s^2$ 4) $7m/s^2$
3. A particle moves such that $S = 3t^3 - 2t$. At what time will the acceleration be equal to 18 unit/sec^2.
   1) 1 sec 2) 2 sec 3) 4 sec 4) 9 sec
4. A particle moves along a straight line such that its displacement (s) at any time (t) is given by $S = t^3 - 6t^2 + 3t + 4$. Where 'S' is in metres and is in seconds. The velocity of the particle when its acceleration zero is
   1) -9m/s 2) 12 m/s 3) -12m/s 4) 9 m/s
5. A particle moves along a line by $s = t^3/3 - 3t^2 + 8t$ then the distance travelled by the particle before it first comes to rest is
   1) 20 2) 20/3 3) 3 4) 60
6. The height 'y' and the distance 'x' along the horizontal, for a body projected in the vertical plane are given by $y = 8t - 5t^2$ and $x = 6t$. What is the initial velocity of the body?
   1) 20 m/s 2) 10 m/s 3) 5 m/s 4) 15 m/s
7. The distance 'x' of a particle moving in a one dimension under the action of a constant force is related to time 't' by equation $t = \sqrt{x} + 3$, where 'x' is in metres and 't' is in seconds. The displacement of the particle when its velocity zero is
   1) 2m 2) 1m 3) 3m 4) zero
8. A car moves along a straight line whose equation is given by $s = 12t + 3t^2 - 2t^3$ where s is in metres and t is in seconds. The velocity of the car at start will be
   1) 16m/s 2) 12 m/s 3) 9m/s 4) 7m/s
9. The co-ordinates of a moving particle at a time t, are given by, $x = 5 \sin 10t$, $y = 5 \cos 10t$. The speed of the particle is :
   1) 25 2) 50 3) 10 4) None
10. A particle starts moving rectilinearly at time t = 0 such that its velocity 'v' changes with time 't' according to the equation $v = t^2 - t$ where t is in seconds and v is in m/s. The time interval for which the particle retards is
    1) $t < 1/2$ 2) $1/2 < t < 1$ 3) $t > 1$ 4) $t < 1/2$ and $t > 1$
11. For a particle moving on a straight line it is observed that the distance x at time t given

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PHYSICS FOUNDATION CLASS-X

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9. A particle is moving along east direction with a velocity v m/s and suddenly turns along north direction with the same velocity v. Then its resultant velocity is

1) 2v 2) $\sqrt{2}$ v 3) $\sqrt{3}$ v 4) $\sqrt{5}$ v

10. A man walks 5 km westwards then 10 km Northwards and again 5km Westwards. The distance and displacement respectively are _____ and _____

1) 15 km, $10\sqrt{2}$ km 2) 20km, $10\sqrt{2}$ km

3) 10km, $20\sqrt{2}$ km 4) 15km, 20 $20\sqrt{2}$ km

11. A man runs along a circular path of radius 14m. When he completes half of the circle, his displacement is

1) 28 π m 2) 14^2m 3) 28m 4) 14m

12. A body travels around a perimeter of a semi circle of radius 'r'. The ratio of the distance traveled to his displacement is

1) 1 : 1 2) $\frac{\pi}{2\sqrt{2}}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{\sqrt{2}}$

13. A body travels towards north with velocity $2ms^{-1}$ for 20s and then moves towards east with velocity $3ms^{-1}$ for 10s. Its average velocity is

1) 5 ms^{-1} 2) $\frac{5}{3}$ ms^{-1} 3) $\frac{3}{5}$ ms^{-1} 4) 2.5 ms^{-1}

14. A goods train traveling at a speed of 20 kmph for one hour stops at a station for one hour and then continues its journey at 40 kmph for 10 hours. The average speed of the train for the total journey is

1) 35 kmph 2) 210 kmph 3) 42 kmph 4) 30 kmph

15. During a journey ' a ' travels x kilometers with velocity ' V ' and the next ' x ' kilometers with double the velocity. The average velocity during this journey is

1) 3V/2 2) 3V 3) $2V^2/3$ 4) 4V/3

16. An athelete completes one round of circullar track of radius R in 40second. What will be the displacement at the end of 2min .20sec.

1) 2R 2) π R 3) 2 π R 4) 6 π R + R

17. A clock has its miniute hand 4.5 Cm long .Average velocity of the tip of the minute hand between 6.00 a.m to 6.15 a.m.

1) $\frac{1}{100\sqrt{2}}$ Cm/sec 2) $\frac{2}{100\sqrt{2}}$ Cm/sec 3) 46.6 Cm/sec 4) $4.5\sqrt{2}Cm/\sec$

18. While driving home from school you travel at 80 km/h for 120 km then slow to 40 km/h. You get home in 2 hours . How far is your hometown from school and what is the average speed?

1) 140km 2) 150Km 3)200Km 4)160Km